Signal Before Sense: An Information-Theoretic Authenticity Assessment of the Voynich Manuscript

How not to be fooled by a problem built to seduce: put every claim past a null model before reading meaning — signal before sense — the method, applied to the Voynich Manuscript.

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ABSTRACT

The Voynich Manuscript (Beinecke MS 408, vellum radiocarbon-dated 1404–1438) is written in an unknown script that has resisted every decipherment attempt for over a century. We set aside decipherment and ask an antecedent, decidable question: does the text exhibit the statistical fingerprints of encoded communication, or those of meaningless fabrication? Communication is substrate-independent — any text that transmits meaning, in any language known or unknown, obeys characteristic laws of frequency, predictability, memory, and topical organization. Applying a standard information-theoretic battery to the full Takahashi transliteration (36,923 tokens), and benchmarking against natural language and two null models, we find that the manuscript is decisively *not* random fabrication, exhibits an anomalously low character-conditional entropy, retains long-range statistical memory, and — most tellingly — its vocabulary is organized by illustrated subject matter at 165σ above chance, a signature that survives when the two scribal "languages" are controlled for. Several of these — the low entropy, the combinatorial word-structure, and this topical organization — are established results, reproduced here independently with a single consistent pipeline and cited at each point; we claim priority for none. The evidence is consistent with a genuine information-bearing artifact (a natural language in a rigid script, or a structured cipher) and inconsistent with unstructured gibberish. Reverse-engineering the structure, we then find the words are not lexical but *combinatorial*: just the fifteen commonest two-glyph beginnings and endings already cover 62% of them (against 18% for Spanish), the small, grammar-coupled settings of a few dials whose physical form — a rotating wheel of parts — is drawn in the manuscript's own cosmological folios. This matches the profile of an *a priori* combinatorial language — the conjecture William Friedman reached; the surface statistics, however, cannot decide whether the code carries meaning or is a mechanical generator, and we show that tie to be unbreakable from the text alone. We further show *why*: the code is formally underdetermined — the positional grammar fixes the form and prunes the readings by orders of magnitude, yet leaves roughly 10^{26} meaning-assignments consistent, and the joint image–text channel that might anchor it returns null at full statistical power. The century of failure is the signature of an underdetermined problem, not of its investigators. This is not a decipherment; it is a characterization of the machine — and a proof of what reading it would take.

> **Author's note.** I make no claim to be right. What follows is an honest effort to get as close as I could to understanding this manuscript, with the help of modern computational tools. If anyone finds an objection I have not considered, I would be grateful to hear it — together we can get a little closer. — C.M.V.

Keywords — Voynich Manuscript · information theory · conditional entropy · quantitative philology · combinatorial morphology · cipher hypothesis · manuscript authenticity

HIGHLIGHTS

2.26	**165σ**	**62%**	**0**
conditional entropy h_2 (bits) — vs 3.04 for Spanish	vocabulary–section specificity above chance	of words built from the 15 commonest two-glyph beginnings & endings — a combinatorial code (Spanish: 18%)	accepted decipherments, to date

fachys · ykal · ar · ataiin · shol · shory · cthres · y · kor · sholdy

Folio 1r, line 1 (EVA transliteration, Takahashi). The script has never been read. This paper does not attempt to. It asks two prior questions: is there a message here at all — and if so, what machine produced it?

1 The question worth asking first

Every announced "solution" to the Voynich Manuscript — and there have been many, most recently in 2017, 2019, 2023, and twice in 2025 — has been rejected by cryptographers and linguists within weeks [1]. The common failure is procedural: they attempt to *read* the text before establishing that it can be read. As of mid-2026 a forecasting market prices the probability of a validated, peer-reviewed decipherment by year end at roughly 3% [2].

We invert the order of operations. Four hypotheses exhaust the space: **(a)** a natural language in an unusual script; **(b)** a natural language enciphered; **(c)** a constructed or artificial language; **(d)** a hoax — meaningless glyphs produced to simulate text. Hypotheses (a)–(c) all entail that the text *carries information*; only (d) denies it. That distinction is measurable without reading a single word.

The premise that makes this possible is that **communication leaves substrate-independent traces**. A message is a message whether written in Latin, Nahuatl, or a cipher no one has cracked; the act of transmitting meaning imposes regularities — Zipfian word frequencies, a bounded per-symbol information rate, long-range correlations, and vocabulary that tracks subject matter. These fingerprints are what we test for. We do not translate the manuscript; we detect whether there is anything to translate.

Beyond the authenticity verdict, we then reverse-engineer the structure: the architecture of a "word", the layers of scribal formatting that must be stripped before any decoding, and the outcome of two natural cribbing attacks on the manuscript's most tractable target — the zodiac labels. We report the cribs that *failed* as first-class results: in a problem this crowded with premature "solutions," knowing precisely which footholds do not exist is as valuable as any that might.

2 Materials

We use the Takahashi transliteration — the first and only complete transcription of the manuscript — as distributed in the Landini–Stolfi interlinear archive in IVTFF format [3]

[4], encoded in the Extensible Voynich Alphabet (EVA). Word tokens are delimited by EVA spaces; alignment placeholders are removed and any token containing an unreadable glyph is discarded. This yields **36,923 tokens** over **9,328 distinct word-types**. Page-level metadata provide two additional labels used below: Currier's scribal "language" (A/B) and the illustration class of each folio (Herbal, Astronomical, Biological, Pharmaceutical, Cosmological, Zodiac, Stars, Text).

3 Methods

All methods are standard and reproducible. On the character stream (words joined by a single space) we compute the **unigram entropy h_1** and the **conditional entropies h_2, h_3** — the average information carried by a glyph given the one or two preceding it, obtained as differences of block entropies. At the word level we fit the **Zipf slope** of the rank–frequency distribution and record the type–token ratio (TTR) and word-length distribution.

To probe memory beyond adjacency we compute the **mutual information $I(d)$** between characters separated by a distance d, for d up to 80. To test whether vocabulary is meaning-bearing we measure **section specificity**: for each sufficiently frequent word we take the Kullback–Leibler divergence of its distribution across illustration classes from the global class distribution, and average over the corpus weighted by frequency, following the information-theoretic keyword approach of Montemurro & Zanette [5]. Significance is assessed against a null in which section labels are permuted across token positions (20 replicates).

Baselines. A natural-language control (Spanish; `Don Quijote`, letters-only, matched token count); a **word-shuffled** Voynich (destroys inter-word order); and an **order-0 random** stream (the manuscript's own character frequencies, all structure removed) — the appearance of genuine noise.

4 Results

The results fall in two movements. The first (§4.1–4.6) establishes that the text carries a message at all; the second (§4.7–4.16) reverse-engineers the mechanism that produced it.

Relation to prior work. Several of the individual measurements below reproduce established results — the low conditional entropy [13], the combinatorial word-grammar [14], the two Currier "languages" [6], the topical vocabulary organization [5], and the self-similar generation signature [8]. We cite these at each point and claim no priority for them; independent reproduction with a single consistent pipeline is itself a check. What is our own is the synthesis, the unified low-repetition/low-entropy criterion applied across languages (§4.4, §5), the minimal-machine control (§5), the subject-recovery classifier (§4.14), and the robustness checks (§4.15).

4.1 It is not random — the word level is organized

The order-0 random stream is what fabrication-as-noise looks like: a near-flat Zipf slope (–0.16), almost every token unique (TTR 0.90), and $h_2 \approx h_1$ (no predictive structure). The manuscript is the opposite on every axis — a Zipf slope of –0.83 (Spanish: –0.96), heavy word reuse (TTR 0.25), and h_2 far below h_1. Whatever the manuscript is, it is **not** someone scattering glyphs at random.

CORPUS	TOKENS	TYPES	TTR	H₁	H₂	H₃	ZIPF	⟨WL
Voynich (all)	**36,923**	**9,328**	**0.25**	**4.09**	**2.26**	**1.84**	**−0.83**	**5**
· Currier A	10,833	4,015	0.37	4.16	2.23	1.74	−0.71	6
· Currier B	26,090	6,539	0.25	4.03	2.16	1.79	−0.84	5
Spanish (Quijote)	36,923	5,825	0.16	3.97	3.04	2.56	−0.96	4
Voynich shuffled	36,923	9,328	0.25	4.09	2.26	1.87	−0.83	5
Random (order-0)	36,911	33,215	0.90	4.09	4.05	3.97	−0.16	5

Table 1. Core battery. The random null diverges from the manuscript on every measure; the manuscript tracks natural language on organization (Zipf, TTR) yet departs sharply on predictability (h_2, h_3).

4.2 The anomaly: glyphs are too predictable

Here the manuscript parts company with natural language — in an unexpected direction. Its conditional entropy h_2 = 2.26 bits sits *well below* Spanish (3.04) and every studied European language (typically 3.0–4.0); by h_3 the gap widens (1.84 vs 2.56). Given one glyph, the next is far more predictable than in any ordinary language. The text is **more** regular than natural writing, not less — the manuscript's most durable and most peculiar property — recognised as its signature anomaly since Bennett (1976) and analysed computationally by Reddy & Knight [13], and reproduced here independently.

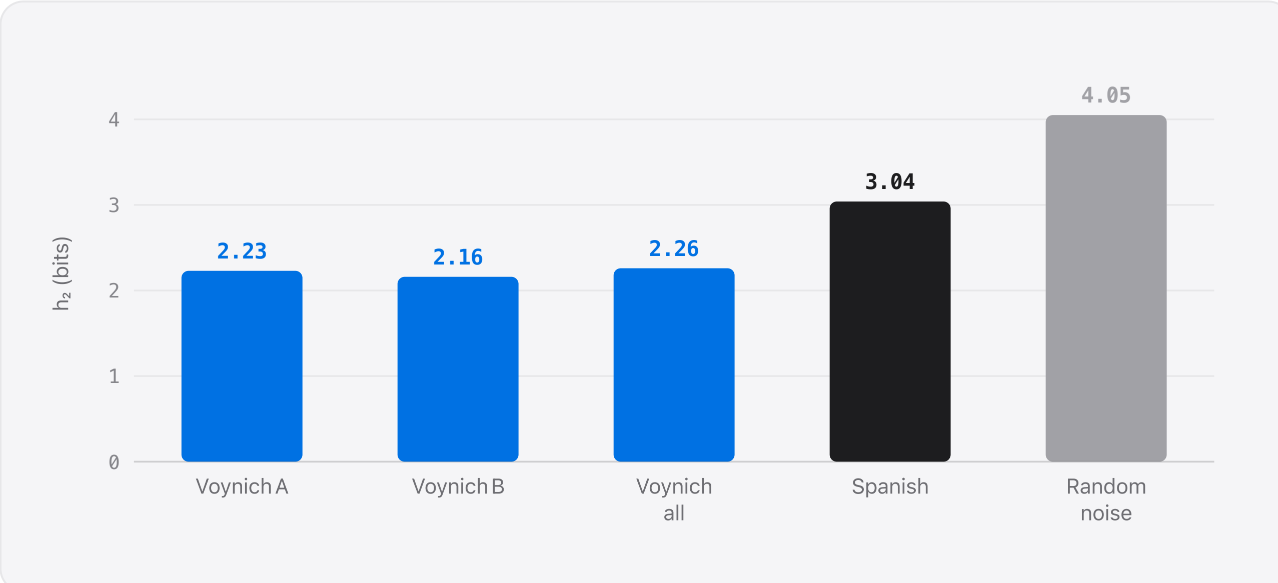

Figure 1. Character-conditional entropy h_2. The manuscript (both scribal languages) is far more predictable than Spanish, and nowhere near the random-noise ceiling. Low h_2 is compatible with a syllabic script or a "verbose" cipher — and also with a highly structured generative process (§5).

4.3 It has memory beyond the word

Mutual information between glyphs decays with distance but remains measurably above the shuffled-noise floor out to $d = 80$ characters — many words apart. A memoryless process (a short Markov chain, or the random null) cannot produce this. The manuscript carries structure that spans well beyond individual words.

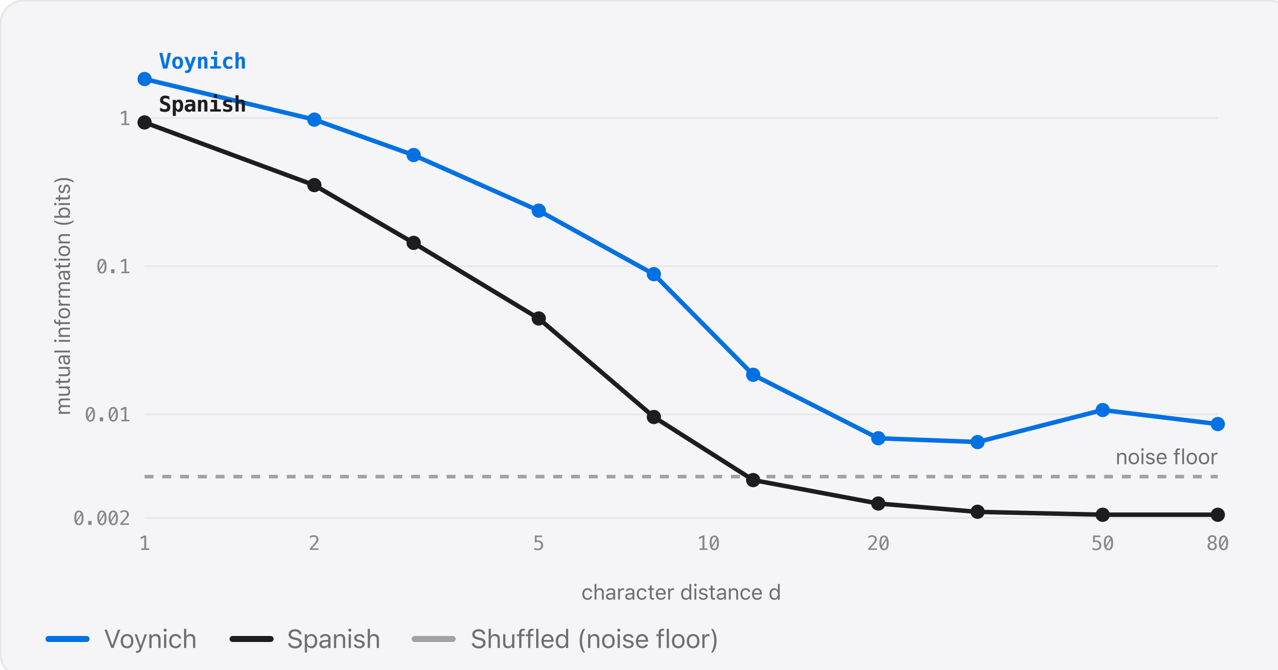

Figure 2. Mutual information I(d) vs. character distance (log–log). Both real texts decay; the shuffled null pins the noise floor. The manuscript stays above that floor at every distance measured.

4.4 Vocabulary tracks subject matter — the strongest result

If the words mean something, they should cluster with the pictures. They do, overwhelmingly. Words are concentrated within their illustration class at **0.574 bits/token** versus a permuted-null expectation of **0.183** — an excess of 0.39 bits, or roughly **165 standard deviations** — reproducing and sharpening the topical vocabulary organization first demonstrated by Montemurro & Zanette [5]; the result here is a re-measurement, not a new discovery. The herbal folios speak a different vocabulary from the "stars/recipes" folios, in exactly the way a real herbal differs from a real almanac.

The obvious objection is that Currier's two scribal languages A and B occupy different sections, so the effect could be dialect, not meaning. It is not: repeating the test *within* a single language holds the signal at 75–77σ (Table 2). Topical vocabulary organization is present inside each scribal system, not merely between them.

PARTITION	TOKENS	SECTIONS	OBSERVED	NULL	EXCESS	Z
Full manuscript	**36,923**	**8**	**0.574**	**0.183**	**+0.391**	**165**
Within Currier B	22,979	5	0.322	0.112	+0.210	77
Within Currier A	10,792	3	0.280	0.086	+0.195	75

Table 2. Section specificity (bits/token) vs. a label-permuted null. The effect survives restriction to a single scribal language, excluding the A/B dialect confound.

4.5 Two systems, not one

Currier's A and B behave as statistically distinct systems: A reuses words less (TTR 0.37 vs 0.25) and runs longer words (6.26 vs 5.46 glyphs). They share the low-entropy signature but differ in their surface statistics — consistent with two scribes, two dialects, or two states of one system, a distinction recently reconfirmed quantitatively [6]. Decomposed into parts, the two share roughly 60% of their top affixes — one machinery — yet each keeps a characteristic accent: A favours the endings `–ol`/`–or`, B the ending `–edy`. The pair reads less as two separate languages than as two settings of a single combinatorial system.

4.6 On reading direction — a caution

A natural instinct is to ask whether the text is being read in the wrong direction. We flag a methodological subtlety: character-conditional entropy is *direction-symmetric by construction* (h_2 forward = h_2 backward = 2.258, an identity of mutual information), so it cannot detect left-to-right versus right-to-left reading. The manuscript's genuine directional signature is **morphological** — glyphs occupy near-obligatory positions within a word (certain forms only begin, others only end it) — an asymmetry that requires position-aware measures, and which independent 2026 work finds to be strong and cipher-like [7]. The directional question is live and unresolved; it simply is not answered by entropy alone.

4.7 The architecture of a word

The low conditional entropy has a concrete cause: words are assembled from near-obligatory **positional slots**. A small set of glyphs opens words (`qo–`, `ch–`, `sh–`, `ok–`, `ot–`) and a small set closes them (`–y` ends 38% of tokens; `–dy`, `–aiin`, `–ol` dominate the rest). A script in which the first and last positions are this restricted is *not a plain alphabet*, where any letter may occupy any position. The unit of the system is not "one glyph, one sound"; the profile fits a syllabary or a **verbose cipher** (one underlying unit spelled by several glyphs), the natural explanation for the entropy anomaly of §4.2.

This positional locking can be quantified, and it decides a specific alternative. Were the glyphs the *digits* of a positional number system — as a numeric or gematria reading requires — any glyph could occupy any position, and glyph identity would carry no information about it. The mutual information between glyph and position-class (initial/medial/final) is **0.72 bits** in the Voynich, against 0.26 for Spanish, 0.15 for Latin, and **0.014** for a size-matched sample of synthetic base-20 numerals. The glyphs are thus *more* position-bound than the letters of a natural language and roughly fifty times more than digits: `q` is word-initial 100% of the time, while `n`, `m` and `y` are word-final in 89–99%. A numeric reading — each glyph a digit, words as numbers — is therefore excluded: the symbols behave as slot-bound morphemes, not as free digits, sharpening the combinatorial picture of §4.13. This value also exceeds every one of nineteen Latin-alphabet languages measured with the same pipeline; the highest, agglutinative Finnish (0.42), is the closest natural analogue — heavy suffix-stacking is the linguistic mechanism that most resembles the manuscript's dials — yet falls far short, and a constructed *a priori*-style language (Esperanto, 0.30) does not approach it either. The Voynich's level is reached or exceeded only by non-alphabetic scripts — logographic Chinese and Japanese (0.8–1.0, with 500+ distinct characters) and the syllabic or abjad Hindi (0.70), Hebrew (0.58) and Tamil (0.53), where each character is a syllable- or consonant-sized unit with a strong positional role — but all achieve it through hundreds of glyphs or much shorter words (Hindi mean length 2.9 vs 5.1 here), where locking is partly mechanical; at comparable inventory and word length the manuscript's positional locking is the most extreme in the sample. The gap is neither a finite-sample nor a word-length artefact: at a common sample size the estimate's shuffle floor is ≈0.00 bits (bias-corrected value 0.72), its 95% bootstrap confidence interval [0.70, 0.74] does not overlap that of the next-highest language (Finnish, [0.41, 0.45]), and restricting every corpus to 4–7-glyph words leaves the Voynich highest (0.74). No natural orthography combines its small inventory (21 glyphs), long words, and this degree of locking — the profile of a templated slot system, consistent with the syllabary-or-verbose-cipher reading above rather than a plain alphabet. (Glyphs are counted as single EVA characters; treating `ch`/`sh` as units would only raise the locking further.)

The positional constraint reaches beyond the word to the line. A word's position within its line carries measurable information about its form: the mutual information between line-position and prefix is 0.19 bits, roughly sixfold above a within-line-shuffled null — concentrated at the line edges (position 0 favours the marker prefixes `d−/s−/y−`, position 1 peaks on `ch−/sh−`). Where a token sits partly governs how it is built, not merely what it is — a further sign that layout, not a sentence, is the organising frame.

4.8 Two layers of scribal formatting must be stripped first

Before any glyph can be assigned a value, two non-linguistic layers have to be removed — the manuscript's equivalent of the message preambles and padding that Bletchley Park stripped before reading Enigma traffic.

Paragraph capitals. The first word of a paragraph begins with a "gallows" glyph (`p, t, k, f` — the tall, ornamented forms) **78%** of the time, versus 6% mid-line. This is an ornamental initial, a drop-cap; its first glyph is decoration, not sound, and should be discounted.

Line-initial markers. Physical lines mid-paragraph favor a different set of openers (`d−, s−, y−`). Strip that first glyph and, **28%** of the time, a common word appears underneath — versus 13% for mid-line words: `daiin→aiin`, `saiin→aiin`, `sol→ol`, `dar→ar`. The line-start glyph is an added marker glued to an ordinary word (the phenomenon earlier catalogued as "Grove words" [15]). Line *ends* carry their own marker: the glyph `m` is a terminal-position ending — it closes its line **71%** of the time (16% base) and reappears at the end of standalone labels (7.4% of zodiac labels versus 2.3% in running text). It is a real, productive ending, attaching to as many distinct stems as the ordinary `−o` (≈353 vs 352 word-types), but a *positional* variant rather than a semantic one: it does not detach like an appended sigil (only 23% of its stems occur bare) and is spread evenly across sections rather than concentrated where the manuscript would mark degrees — so not, for instance, a degree marker. By contrast –n itself is position-neutral (14.5%). Line lengths are too irregular for strict metre (CV ≈ 0.45), so this is line-level formatting, not verse.

4.9 Two cribs on the zodiac, and why both fail

The zodiac folios are the manuscript's most tractable target: each medallion carries a ring of **~30 labelled nymphs** — a calendrical scaffold (the 30 degrees or days of a sign) that supplies the kind of known, ordered structure Bletchley exploited in stereotyped message openings. Two natural cribs follow, and the data refute both.

Crib 1 — labels as recurring day/degree numbers. If the ~30 labels enumerate positions, they should recur, in correlated order, across the twelve medallions. They do not: **267 of 296 labels are unique** (only 17% appear more than once). The labels are not a recycled numbering.

Crib 2 — labels as names the surrounding text repeats. If a nymph's label names something, its stem should surface in the same folio's ring text. Overlap between a folio's labels and its own body text runs at just **1.18×** the cross-folio control (1.08× on stripped stems) — statistical noise. There is no measurable label-to-text co-reference.

Both footholds that "should" exist are absent — a compact illustration of why the manuscript remains unbroken, and a boundary any future proposal must respect.

4.10 Section identity is morphological, not lexical

The 165σ section signal of §4.4 is real, but locating *what* distinguishes the sections is sobering: it is the distribution of **affixes**, not a vocabulary of content words. No isolable lexeme for "water", "root", or "star" emerges; instead each section prefers a family of prefixes and suffixes. Concretely, each section carries a signature prefix (herbal `ch–`, biological `qok–`, zodiac `ot–`), and even the optional `q–` marker swings from 25% of biological words to 1% of zodiac ones; the prefix predicts the section far above chance (mutual information 0.093 bits, $z \approx 240$ against a section-shuffled null). The affixes are also where the surface vocabulary comes from: stripping them collapses the 6,300 word-types of this affix-analysis subset onto ~2,100 cores, and removing the prefix alone merges 41% of types (of 703 `q–` words, 61% have their bare form also attested — the marker is largely optional). The affixes are few and largely domain-general — roughly a dozen effective prefixes and a dozen suffixes, each section merely favouring some — so the "filler" is a systematic, section-shifted layer rather than noise; the section is legible from both the affix mix and the larger core inventory (the frequent cores `e`, `o`, `a` are thin, but rarer cores are section-bound). The suffixes' section-profiles are moreover close to one-dimensional (87% of variance on one axis), an ordinal-scale signature — consistent with, though not proof of, a small graded "quality" dial in that slot — an exploratory observation, not a confirmed result. This does not decide meaning: a per-section classifier morpheme and a per-section-tuned generator both produce section-dependent affixes (§5).

Two further measurements locate and characterise this section structure. First, within the word the section signal sits mainly in the *core*: the core discriminates section 2.3× more strongly than either affix (excess mutual information 0.19 versus 0.08 bits) — the variable slot, not the frame, carries most of the 165σ effect. Second, the frame is not held fixed across sections but re-tunes *in step with the content*: across the five sections the distance between their affix profiles tracks the distance between their core vocabularies at **r = 0.91** (permutation p = 0.011), so sections whose contents are most alike (herbal and pharmaceutical) also share the most machinery, while the most dissimilar (biological and astronomical) share the least. The manuscript reads as one coherent system re-tuned smoothly across related topics, not a set of independent per-section generators. (Topic and scribal hand — Currier A versus B — are confounded across these sections, so "thematic" relatedness cannot be cleanly separated from shared scribal habit; and a section-discriminating core is equally consistent with a thematic catalogue and a section-specific core generator.)

SECTION	CHARACTERISTIC WORDS	PATTERN
Herbal	`daiin, chol, chor, shol, shor`	ch-/sh- + -ol/-or
Stars / recipes	`qokeey, okeey, oteedy, otaiin`	ok-/ot-/qok- + -eey/-aiin
Biological	`shedy, qokedy, qokain, qotedy`	qok-/sh- + -edy

Table 3. Section-defining words (running text, ≥50% concentrated in-section). What separates the sections is which affix families dominate — the topical signal is carried morphologically, not by a decodable content lexicon.

4.11 Not linear prose: the line is a closed unit

Two final measurements bear on the tacit assumption that the manuscript is continuous prose to be read across wrapping lines — an assumption that encodes our own reading conventions, not a universal.

It does not repeat like a procedure. A recipe or procedural text is formulaic: it reuses instruction templates constantly. Trigram (three-word) repetition in Spanish prose runs at **9.6%**; across every Voynich section it sits between **0.0% and 0.9%** — one to two orders of magnitude *lower*. Nearly every three-word sequence in the manuscript is unique. Whatever it is, **it is not the repetitive list of procedures a recipe book or al-**

manac would be — the single most popular reading of the manuscript, and one this statistic directly undercuts.

Its lines do not flow into one another. Mutual information between adjacent characters *within* a line is 1.72 bits; across a line boundary it collapses to **0.29 bits** — a sixfold drop. In wrapping prose, the last word of a line predicts the first of the next; here that link is nearly severed. The line behaves as a self-contained unit — a row or entry — rather than a fragment of a sentence continuing overleaf. Together with the manuscript's many radial and circular text layouts, which are not read left-to-right at all, this argues that imposing a linear reading order is itself a mistake.

Within a line, though, words are linked. Consecutive words are not independent: how one ends predicts how the next begins. The mutual information between a word's suffix and the following word's prefix is **0.11 bits**, and it survives the two obvious confounds — holding at $z \approx 110$ against a null that shuffles only *within* each section (so it is not shared topic), and at $z \approx 88$ after excluding near-copy neighbours (so it is not the local self-similarity of §4.13). A memoryless whole-word generator would not produce this coupling; the manuscript is therefore *at least first-order* at the level of its affixes — a local linking rule. Like every result here it constrains the mechanism without deciding it: a first-order generator carries this structure as readily as a meaningful text would.

4.12 A spatial grammar binds text to image

The transliteration encodes not only the glyphs but the *kind and placement* of every text block — running paragraph, label beside a figure, circular ring, radial spoke. Mapping these per page reveals that text is not laid down independently of the illustrations: where figures appear, the text arranges itself around them in an illustration-type-specific pattern.

The binding is tight enough that layout alone identifies content. Every zodiac folio (12/12) carries central circular text ringed by nymph labels; the pattern "circular text + labels" occurs on the zodiac pages and **nowhere else**. "Circular text + paragraph" is exclusively cosmological. Pharmaceutical and biological pages pair labels with paragraphs (labelled jars and organs); herbal and recipe pages, lacking such figures, are plain paragraphs. In short, the position of the writing is dictated by the structure of the picture — the text behaves as annotation organised by the image, not as a self-standing document the images happen to accompany. This is direct, transliteration-level support for the image-as-primary-channel reading of §5, obtained without any pixel analysis.

4.13 The vocabulary is combinatorial

A final property unifies the structural results. The manuscript's words are not drawn freely from a lexicon; they are assembled combinatorially from small inventories — the word-grammar mapped in detail by Stolfi's "crust–core–mantle" model and by Reddy & Knight [14][13]. A short list of prefixes and suffixes decomposes **81%** of tokens into prefix + core + suffix, and just **15 prefixes and 17 suffixes**, with a modest core set, generate 90% of the running text. The scheme is templated to a degree natural language is not: with only the fifteen commonest two-glyph beginnings and endings, **62%** of Voynich tokens are covered — against **18%** for a matched Spanish sample. The residual cores are thin (single glyphs dominate), so the information a word carries sits in its affixes — its category and state "dials" — not a rich stem. At the level that matters — the affix dial itself — the grammar is permissive, not sparse: of the **255** possible prefix×suffix frames (fifteen prefixes × seventeen suffixes), **94%** actually occur, only ~6% never appearing. (Whole-word combinations look far sparser, but that is an artefact of many rare single-use cores inflating the denominator, not a wall of forbidden forms — consistent with the graded, no-hard-prohibition coupling below.) The same fact seen from the other side is stark: whereas three-word sequences almost never recur (0.4%), the sub-word atoms recur relentlessly — **69%** of atom-trigrams repeat — so it is the parts, not the words, that are the reused units of the system. That the parts repeat while the word-sequences do not is, moreover, specific to the manuscript rather than a generic property of text: in a size-matched Spanish sample the two levels rise together (three-word sequences recur at 16%, a part-to-word repetition ratio near 6×), whereas the Voynich holds word-sequence repetition near zero while its parts saturate — a ratio near 140×, some twenty times higher. The reused unit is demonstrably sub-word, not lexical.

This is the profile of a combinatorial generation system — of the kind Ramon Llull's revolving concentric wheels (*Ars Magna*, 13th–15th c.) were designed to produce, and which the manuscript's own circular and radial text layouts (§4.12) physically resemble. A Voynich "word" then reads less as a spoken form than as a *setting of the wheels*. Whether such a system carries meaning (a classificatory code) or only structure (a mechanical generator — the mechanism behind the Cardan-grille hoax hypothesis [9]) is the fork §5 leaves open; the section-semantic signal of §4.4 tilts it toward the former.

Reconstructing the combination table sharpens this. Prefix and suffix are *not* independent: their mutual information (0.32 bits) is some twenty-four times an independence null, so the dials do not spin freely — a soft grammar couples them, as a designed code would and a free generator would not. The couplings are graded preferences, not hard prohibitions (no expected-but-absent pairs appear). And the dozen surface prefixes collapse, by their suffix behaviour, into only three or four equivalence classes (`ch/sh`; the gallows family `qok/ok/ot/k`; `o/y`; `d`) — the underlying category dial is small. Fittingly, the cosmological folios are drawn as literal wheels: concentric rings and some thirty radial spokes carrying text inward toward a central rosette (e.g. f69r), a *rota* in the precise sense of Llull's figures — the mechanism and its physical form appearing on the same page. Closer inspection of that hub is telling: around a central star sit a handful of single glyphs (`d, o, y, s, l` — forms that resemble numerals only because the script's letters do) — precisely the atoms from which the surrounding words are built. The wheel carries its generating alphabet at its axis and its products at its rim.

The setting drifts — one dial, not the other. If a word is a setting of the wheel, the setting need not be fixed: it can advance across the manuscript. It does — but only on one dial. Comparing each herbal folio's affix profile to every other against their separation in the manuscript, the *suffix* inventory drifts steadily (neighbouring folios share it, distant folios diverge: $r = -0.10$ against a permuted null, an effect that survives restriction to a single Currier language, $z = -4.3$), while the *prefix* inventory stays flat (no significant drift). A generic cause — an evolving scribal hand, a gradual change of topic — would move both dials; that only the suffix drifts points to a structured, dial-specific setting rather than mere authorial drift. The plant drawings drift too, but on a separate clock: image morphology and suffix profile each track position in the book, yet neither predicts the other once position is held fixed (partial $r \approx -0.03$). The text's slowly-turning dial and the pictures' slow change are two independent processes — the same text–image autonomy of §4.12 and §5, now visible in the dynamics. (We report this as a minor structural regularity, not proof of a mechanical rotor; drift is consistent with a stepping code and with a slowly-varying scribal habit alike.)

Interactive companion. Because the claim is about a wheel, it can be turned. An interactive reconstruction of f69r places the manuscript's own transliterated tokens on rotating rings; turning them shows which alignments decompose into a licensed prefix·-core·suffix form or match a word attested elsewhere on the folio. It is an instrument for probing the wheel hypothesis, not a claimed decipherment.

4.14 The classification table is partially recoverable

If the words are settings of a classificatory wheel, the wheel positions should encode what each page is about — and they do, recoverably. A folio's illustration-class can be predicted from its *affix profile alone* (nearest-centroid, leave-one-out across 207 folios, no lexical content used) at **82%** accuracy against a 62% majority baseline — a twenty-point gain from morphology only (71–82% depending on whether the affix inventory is curated or purely data-driven; the effect is robust, the exact figure is not — §4.15). The diagnostic markers form a first-level classification table: herbal folios are flagged by the prefixes `cth-/ckh-`, biological folios by `qol-` (4.1× enrichment) and the suffix `-edy`, pharmaceutical by `-ol/-ody`, the zodiac by `ot-`, the astronomical by `-s/-eey`. This bears directly on §5's fork: a content-free generator has no access to a page's subject and cannot align its dials to it, so an 82% prediction of the subject from the affixes alone is the strongest single indication that the combinatorial code carries *classificatory meaning* rather than none. What stays unrecovered is the item level — which plant, which star — carried by the thin core dial, whose reading requires the figure-to-code anchor that §5 shows to be empirically null. We recover the table's top row (the subject), not its entries — the exact reach, and limit, of an *a priori* classification approached from the text alone.

4.15 Robustness

Three checks guard against artefacts of transcription and method. **(i) Transcription:** repeating the core measurements on three independent transcriptions — Takahashi, the First Study Group, and Currier — gives near-identical values (combinatorial coverage 64–65%, trigram repetition 0.1–0.2%, conditional entropy h_2 2.24–2.27), so no result rests on a single reading. **(ii) Affix choice:** replacing our curated prefix/suffix inventory with a purely data-driven one (the twenty most frequent two-glyph beginnings and endings) preserves the combinatorial coverage (64% versus 18% for Spanish) and the subject-recovery effect, though the classifier falls from 82% to 71% — the effect survives, the exact figure does not. **(iii) Genre:** a control "glossary" built from a real language's unique word-types matches the manuscript's near-zero phrase repetition (0.0%) but not its depressed entropy (3.55 versus 2.26), confirming that the manuscript behaves as a list of *generated*, not real, words. **(iv) Alternatives tested and rejected:** two further accounts of the low-entropy, templated profile fail directly. Heavy medieval abbreviation of Latin does not reproduce it — abbreviating a Latin sample *raises* its conditional entropy (3.50→3.55) and *lowers* its templating (16%→9% two-glyph coverage), moving away from the manuscript rather than toward it, because abbreviation adds sign- and form-variety. And the third of tokens that break the slot grammar are not a hidden content layer: they carry no more section-specific signal than the conforming majority (weighted KL 0.372 vs 0.370 above a shuffled floor). Third, the mechanism's advance is not *numeric*: were the dials stepped by a counter or rotor, the token sequence would show periodicity, a dominant non-zero step, or a modular positional cycle — and all three come back null (autocorrelation $z \approx 0$ at lags ≥ 2; the modal inter-word step is zero, i.e. repetition rather than counting; mutual information with position mod k never exceeds 0.01 bits). The couplings that link the dials are associative, not arithmetic. A final control bears on the syllabary reading: even a maximally slot-syllabic script — Tibetan, whose orthography assembles each syllable from positional consonant slots — fails to reproduce the profile, returning the *highest* conditional entropy and the *highest* phrase-repetition of any corpus we measured, the opposite of the manuscript on both axes; the low-entropy, non-repeating anomaly is not a by-product of slot-based writing.

4.16 The rule space, and why the code is undecidable from the text

The combinatorial machine of §4.13 has a definite size, and that size is telling. Thirteen effective prefixes, roughly 147 cores and eleven suffixes generate about **21,000** possible words, of which **7,494** are attested — a third of the space filled. This is the scale of a controlled catalogue, not of an open language. Within the word the labour is divided: the affix slots draw on a tiny inventory (~13 and ~11 values) and behave as a *categorical frame*, while the core is the high-cardinality, most graded slot (about 55 distinct values per folio against ~15 for either affix), the variable that carries the specific *value*. The division reaches individual glyphs. Some are purely positional markers — `q` is word-initial 100% of the time, `m` and `y` close words 86–93% of the time, `h` occurs only medially, inside digraphs — the structural or relational scaffolding, the manuscript's equivalent of a word's function-words and punctuation; freer glyphs such as `o`, `c` and `l` carry the value. A word thus reads as a filled record: a categorical frame with a graded value slotted into it.

How many codebooks could underlie such a system? Naively the number is astronomical — a bare glyph-to-letter substitution over the 26-glyph inventory already admits $26! \approx 4\times10^{26}$ mappings. But that figure overcounts, because the positional grammar is itself a set of hard rules legible straight from the text: **56%** of the 676 possible glyph bigrams and **92%** of the 17,576 possible trigrams never occur. These prohibitions collapse the space of valid word-forms roughly **1,600-fold** — from the ~12 million strings a free 26-symbol alphabet of this length would allow to the ~7,500 attested — and fix each glyph's structural role. Inferring that grammar is real progress, and it is how a cipher is normally broken: not by brute-forcing the factorial but by pruning it with constraints.

The pruning, however, constrains form, not meaning. Relabelling the value slots — deciding that a given core denotes one plant rather than another — leaves every attested form unchanged, so no internal evidence can separate the surviving assignments. Decoding is blocked not by the raw size of the space, which the grammar tames, but by the absence of an *external anchor*: a crib, a single figure or object whose text is independently known. The one channel that could supply it — the drawings — returns null (§5), and that null is not for want of statistical power: a synthetic test shows that a visible-feature-to-word rule present in as few as 20–30% of the plants would be detected in ≥90% of trials, yet the real data yield nothing. What the text does fix is structure and topic — a folio's subject is recoverable from its affixes at 71–82% (§4.14). What it cannot fix is the referent. In this precise sense the manuscript is a finite combinatorial catalogue whose form is fully legible and whose codebook is, from the text alone, formally underdetermined.

5 Discussion

What is ruled out. The naïve hoax — glyphs scrawled to look like writing — is excluded decisively. Genuine noise shows a flat Zipf slope, near-total type uniqueness, and no conditional or long-range structure; the manuscript shows the opposite on all four. Something here is organized to carry information.

What remains open. The low conditional entropy is a double-edged result. It is naturally explained by an information-bearing system — a syllabary, an abjad, or a "verbose" cipher in which one plaintext unit maps to several glyphs. But a sophisticated *mechanical* hoax can also produce it: Timm & Schinner showed that a self-citation copying procedure reproduces several Voynich statistics, including the low entropy [8]. Local-entropy evidence alone cannot separate "real language in a rigid script" from "structured artificial generation."

Where the balance tips. The section-specificity result (§4.4) is the hardest for a content-free generator to fake, because a context-free copying process has no reason to align vocabulary with the illustration on the page — yet the manuscript does so at 165σ, and within each scribal language besides. Coupled with long-range memory, this pushes the weight of evidence toward a system that encodes topical content. Our findings sit squarely with the direction the field itself moved in 2026 — toward layered, positional, cipher-like structure [7][10] — rather than toward the hoax verdict.

A named mechanism, and a striking precedent. The combinatorial profile of §4.13 has historical form. Ramon Llull's *Ars Magna* (c. 1305) generated statements by rotating concentric lettered discs — volvelles — to combine elements from fixed inventories, a technique alive in the fifteenth-century Europe of the manuscript's own vellum. Two lineages descend from it. One is cryptographic: Alberti's rotating cipher disk (1467), then Trithemius, Vigenère, and ultimately the rotor of the Enigma machine — each a wheel that recombines alphabets [11]. The other is linguistic: the a priori "philosophical languages" (Kircher — through whose hands the manuscript itself passed — then Wilkins, Dalgarno, Leibniz), in which a word is not a sound but a code assembled from category, sub-category and modifier slots. Our decomposition — prefix as category, suffix as state, thin core as item — is precisely the morphology of such a language, and the manuscript's circular text pages are what its generating wheels would look like. This is, strikingly, the conclusion reached independently by **William F. Friedman** — the cryptologist who broke Japan's PURPLE cipher — who after decades judged the Voynich "an early attempt to construct an artificial or universal language of the a priori type" [12]. Our statistics recover his verdict from the text's own structure. It reframes the task: an a priori language is not translated but *reconstructed* — one recovers its classification table, not a phonetic alphabet — which is why the word-hunting cribs of §4.9 fail by design.

Consistency with origin. The vellum's radiocarbon date (1404–1438) and the manuscript's provenance place it in fifteenth-century northern Italy and central Europe — the very milieu in which Llull's combinatorics circulated. A structural test once seemed to bear on the source language, but on scrutiny it does not, and we withdraw it. A recurring proposal holds that the script conceals a Semitic abjad; we had read the round glyphs (`a, o, e, i, y`) — if vowels, ~43% of the text — as European rather than Semitic. That inference fails on two counts. First, that these glyphs are vowels is unproven: an unsupervised vowel/consonant test (Sukhotin's algorithm), which cleanly recovers the vowels of Spanish and Latin controls, yields *no* clean partition here — the script may not be alphabetic-phonetic at all. Second, the premise that a written abjad carries almost no vowels is wrong: the *matres lectionis* of Arabic and Hebrew already run ~25–30% of letters, and fully vocalised classical Arabic ~40–56% — straddling the manuscript's figure. Vowel density thus neither confirms a European vocalic system nor excludes a Semitic source. The radiocarbon date and provenance remain consistent with the European milieu of Llull's combinatorics, but do not, by themselves, fix the underlying language.

Is it word-based at all? The structural results (§4.7–4.12) sharpen the question in an uncomfortable direction. At the *word* level the manuscript is language-like — Zipfian, heavily reused. But below the word it behaves like a *notation*: rigid positional slots, register-dependent prefixes (labels favour `ot-/ok-`, prose `qo-/ch-`), section identity carried by affix distribution rather than content words, and labels that resist both cribs. This is the profile of a system sitting **between language and notation** — and it invites a hypothesis the decipherment tradition rarely entertains: that the glyph-strings may not encode phonetic words in the way translators assume, but a structured annotation whose meaning is carried positionally and *in concert with the illustrations*. The manuscript's undisputed calendrical scaffolding — twelve zodiac medallions of thirty labelled nymphs, $12 \times 30 = 360$ — is the clearest case where the picture, not the text, is the primary channel. If that reading is even partly right, the century of failure is unsurprising: the cribs fail because they hunt for words in a document that may not be spelling them. Testing this claim requires joint image–text analysis beyond the transliteration used here, and is the direction we regard as most promising.

Two hypotheses that resist separation. The live alternatives — a meaning-bearing combinatorial code (Friedman's a priori language) and a meaning-free mechanical generator (the self-citation model of Timm & Schinner) — turn out to share their surface statistics. Words near one another are modestly more similar than chance (17% lie within one edit of a preceding neighbour, versus 10% once the text is shuffled), the very signature Timm & Schinner read as mechanical copying; yet a small-inventory combinatorial code produces those same local near-variants as an automatic by-product, and the rate barely exceeds natural Spanish (15%). The two accounts are one observation seen two ways — which is why a century of analysis has not chosen between them. Only the global section-semantic signal (§4.4), which a context-blind generator should not produce, tips the balance, weakly, toward meaning. A decisive separation is not available from the text alone; it requires the figure-to-code link that joint image analysis would supply.

Generation is not inscription. Two independent layers must be kept apart, and conflating them is a recurring error. Our statistics probe the *generation* — the process that produced the token sequence, whatever it was. Codicology probes the *inscription* — the physical act of writing. The hands are fluent and confident, with few corrections, in every section; but fluency belongs to the inscription and says nothing about the generation. A scribe transcribes fluently from any source — a memorised cipher, a rotating device, a meaningless exemplar — so the confident ductus neither excludes a laborious encoding (whose labour, if any, preceded the writing) nor implies meaning; equally, our combinatorial and topical results characterise the generation and are silent on whether the writing was original composition or fluent copying. This is why fluency cannot be read, as it sometimes is, as evidence for the hoax verdict or against a cipher: it is the wrong layer. The two channels are complementary and must be analysed apart — and it is the inscription channel we do *not* access here (stroke order, corrections, ink sequence, quire construction) that is most likely to bear on the mechanism, since a physical generating device leaves physical traces a free hand does not.

What the profile most resembles — exploratory interpretation, not a result. The measurements bound the object without naming it; the resemblances below are offered as characterisation, and none is a demonstrated claim. Statistically the text is closest to the *output of a combinatorial slot-generator*: words assembled from prefix–core–suffix positions, a flat token grid rather than a nested tree (§4), and a first-order machine that reproduces the anomalously low entropy. Its nearest historical relatives are the *a priori* philosophical languages, whose category-composed words would yield exactly this slot regularity and per-section topicality (Friedman's conjecture, ref. 12), and the combinatorial divination devices of the Llull lineage (§4.13). What no analogy earns is meaning: no slot decomposition we tried recovers a stable sense, so "resembles a category-generator" describes the *shape of the mechanism*, not a reading. The whole is best pictured as an encyclopaedia in costume — dressed as herbal, astrological and balneological knowledge, its text behaving as topically-biased generated tokens rather than prose. At the level of genre the fit is an *inventory* rather than a narrative: the entries are list-like, not sentential (§4.11), templated from a small set of class-marking prefixes over a large stock of cores — the shape of a catalogue of coded [category][value] entries — and they share a word-list's near-zero phrase repetition (§4.15), while the low entropy marks the entries as generated rather than natural words. This refines the genre without deciding meaning: a genuine catalogue in a constructed notation and a fabricated one would share it. The most historically apt guess for the small ordinal "quality" dial (§4.10) is a Galenic complexional grade — hot–warm–cold, the standard rating of a fifteenth-century herbal — a single ordinal axis matching the largely one-dimensional grade space we measure. We flag this as the leading content hypothesis, not a result, because it cannot be verified: complexion is an attributed, non-visible quality (a drawing, however conventional its parts, does not display "hotness"), and the one test that could anchor it — do plants drawn as hot carry the hot grade — is the figure-to-code link, which returns null three times over, with the drawn plants themselves chimeric and unidentifiable. The label the evidence best fits is one the evidence cannot confirm. Which live reading produced it — invented language, believed oracle, or deliberate hoax — is, as argued above, not decidable from the artefact, because all three yield the same tokens and the same fluent hand.

A constructive test reaches the same verdict. Positing an explicit operating logic — three dials (category, item, state) set to a per-register bias, read as prefix+core+suffix and advanced under the measured grammar — reproduces the word-length distribution almost exactly and the Zipf slope and entropy approximately. Matching the manuscript's repetition, though, requires copying at the level of *whole words*, not parts: part-level recombination leaves the text too varied (type–token ratio 0.39 vs 0.25), while whole-word reuse brings it close (0.30). Yet this does not decide the fork either — natural language reuses whole words still more heavily (Spanish, 0.16) — and while a part-level generator leaves a character-entropy residual (h_2 2.68 vs 2.23), emitting whole words i.i.d. from the observed inventory closes it exactly (h_2 2.26): the residual simply reflects that the words behave as fixed units, not free part-combinations. Yet even a generator that reproduces the manuscript's full fingerprint cannot tell a meaning-bearing code from a mechanical one on surface statistics alone. The tie is, on this evidence, unbreakable from the text; the image must break it.

The mechanical account holds out of sample — with one honest gap. That constructive test is not merely a re-description of the data it was fit on. Trained on half the folios and used to generate the other half, a part-generator carrying the measured couplings — prefix conditioned on the previous word's suffix (§4.11), core on its prefix–suffix frame, suffix on its prefix — reproduces the held-out entropy (2.11 bits), templating (69%) and inter-word link of folios it never saw: the account survives out of sample, not only in fit. But the same exercise exposes a residual no simple mechanism we built closes at once. The manuscript is *at the same time* productive — 58% of the word-types in one half are absent from the other — and more regular than any finite-order model: the coupled generator matches the entropy only by reusing the seen inventory (0% novel words), while a character-level core generator invents new words freely but overshoots the entropy (2.56 vs 2.11 bits, and unchanged from a second- to a fifth-order core). Reproducing both the productivity and the low entropy together would demand a core-formation rule tighter than a fifth-order Markov chain — itself mechanical, but one we do not fully specify. We record this as an open sub-problem, not a result; and, like the rest, it does not tip the fork — a tightly constrained productive grammar is as available to a meaningful code as to a mechanical one.

One mechanism is excluded, however: pure transposition. A transposition cipher reorders units without altering them, so it preserves the text's unit inventory and frequencies. The manuscript's parts are too few and too concentrated for that to yield its statistics: even a full random shuffle of its atoms — transposition taken to the limit — leaves 47% of atom-trigrams repeating, against roughly 10% for natural language. No rearrangement of a natural language can reach these figures, because the excess repetition lives in the small inventory, not the order. Whatever the mechanism, it is *substitutive* — a verbose code — not a scrambling of an otherwise ordinary text.

A convergence worth pursuing. The herbal's plants have long resisted identification because so many are composite — the root of one species, the leaves of another, a third plant's flower. If that assembly is design rather than carelessness, the images obey the same principle as the text: a plant is built from interchangeable parts exactly as a word is built from prefix, core and suffix. Should the two inventories correspond — a root-form to a prefix, a leaf-form to a core — the manuscript would be one combinatorial system expressed in two channels, and its labelled plant-parts (densest in the pharmaceutical section, where each drawn fragment carries a caption) would be the natural bilingual key. Testing it demands systematic annotation of part-morphology against part-orthography — the joint image–text analysis this study repeatedly reaches for. We have since carried out exactly this test, and report it plainly: a frontier vision model annotated the morphology of 114 herbal plants, and separately the labelled pharmaceutical parts, yet neither the discriminating words nor the part-captions predict a shared visible feature above a permuted null — three independent figure-to-code tests, all null. Two facts explain the failure and temper the hope: the drawings are ~61% anatomically chimeric (no stable species referent to map onto), and text and image, though each drifts slowly through the book, do so decoupled (partial correlation ≈ 0). The bilingual key, if one exists, is not in the visible morphology.

Candidates for that crib. The domain analysis that maps layout to content also isolates the words most likely to carry it. Sorting the vocabulary by how concentrated each word is within a thematic domain separates probable content words from the grammatical "glue" that spreads evenly across sections. The botanical folios are led by `daiin` and `chol` (67–75% in-domain, and overwhelmingly mid-line, so genuine content rather than positional markers); the balneological section by `shedy`; while `ol`, `aiin` and their kin spread across every domain, as function words would. Crucially, the concentration itself decides one long-standing question: a universal "element" would appear evenly everywhere, but these words are domain-skewed — `daiin` denotes something botanical, not elemental. None can yet be assigned a specific meaning — that is the 97%-false-positive trap of §6 — but they are the precise tokens a figure-to-code test should target first: if `daiin` names a plant-part, the plants it labels should share that part.

6 Limitations

One transcription (Takahashi) anchors the analysis; its core measurements are reproduced on two further independent transcriptions (First Study Group, Currier) in §4.15, but glyph-boundary decisions in EVA remain interpretive and propagate into the statistics. Mutual-information estimates are positively biased at finite sample size — mitigated here by using one estimator across all corpora and anchoring to a shuffled floor, but absolute values should be read comparatively, not literally. Section labels derive from illustration class, a proxy for topic. And the central caveat: **nothing here decodes the manuscript.** We establish that the text behaves like something with a message; we do not recover the message. A warning follows directly from the combinatorial structure, for any would-be decoder: because the text is assembled from a small stock of recurring parts, freely recombining sub-word units will "find" real words in almost anything — in our trials, six random syllables yield a genuine Spanish word 97% of the time by cherry-picking. Isolated recovered words are therefore never evidence; only a single systematic rule that yields connected, coherent text can be. A concrete instance: matching the manuscript's own affixes against words for "hot", "warm", "cold" and "plant" across twelve languages yields eight apparent hits — several strikingly apt (`chedy`~*chaud* "hot", `chor`~*psychros* "cold", `otaiin`~*botane* "plant") — yet random strings of the same lengths yield 6.3 ± 3.2 such hits, so the matches are statistically indistinguishable from chance ($z = +0.5$, $p = 0.32$). The most seductive cross-linguistic coincidences survive no null at all. This is the trap that has consumed a century of proposed solutions.

Our own empty machine translates just as well. Sharper still: the meaningless generator of §5 — which reproduces the manuscript's full structure yet, by construction, carries no meaning — "translates" at the very same rate as the real text. Matching the most frequent words of each against eight concepts in twenty-two languages (Latin and its Romance variants, plus the agglutinative and Semitic tongues structurally closest to the script), 54% of the manuscript's words land on a real word and the empty machine yields 53% ± 1.0 over forty runs — the manuscript sits at $z = +1.1$, statistically indistinguishable from a device we know to be hollow. And the pattern shows precisely why adding languages never helps: the more one adds, the more everything "translates" alike, until even purely random glyphs match 54% of the time. Nor does picking the best-scoring language rescue the method: rank the candidate tongues by how many of the manuscript's words each "generates" and the ranking is identical to the one the empty machine produces (Spearman $r = +1.00$) — the very same language "wins" for a text we know to be hollow, so the winner reflects which dictionary is easiest to match, not which language underlies the script. Translatability saturates as an artefact of the short-string space; meaning plays no part. It is the century-old failure mode made concrete — the reason "I tried more languages, especially the similar ones" has never once produced a decipherment.

7 Conclusion

This work establishes what kind of object the Voynich Manuscript is, and why it has resisted a century of readings. The text is not empty: it organizes its words like a language, keeps statistical memory across long spans, and aligns its vocabulary with its illustrated subject far beyond chance — even within a single scribal hand. The naïve-hoax hypothesis is *excluded*, not merely doubted. Reverse-engineering the structure then settles what the object is: not an open language but a **finite combinatorial catalogue** of some 21,000 possible records, each a categorical frame filled with a graded value, assembled like the settings of a small combinatorial wheel in Llull's tradition (§4.13) — a machine whose physical form the manuscript's own circular folios draw. Its subject is recoverable: a folio's topic can be predicted from its affixes alone at 71–82%

out of sample (§4.14). And the central result is a proof of limits: the code is **undecidable from the text**. The positional grammar fixes the form and prunes the space of readings by more than three orders of magnitude, yet leaves the referent formally underdetermined among roughly 10^{26} meaning-assignments that no internal evidence can separate (§4.16); the joint image–text channel that might have anchored it returns null, at full statistical power (§5). The century of failure is therefore not accident or want of ingenuity — it is the signature of an underdetermined problem, and we quantify exactly how underdetermined. What the text cannot supply, one channel still could: the codicological record — stroke order, ink layering, correction — for which we set out concrete predictions. We did not read the manuscript. We established, with precision, what it is, why it has not been read, and what it would take to read it.

Appendix A · Transcription fidelity: a pixel-level check across three folios

The entire analysis rests on the Takahashi transliteration (§2), and §6 flags glyph-boundary decisions in EVA as the residual interpretive layer. To gauge how much weight that caveat must bear, we retrieved the Beinecke facsimile of folio f2r — the "cornflower" herbal page — and compared its opening paragraph glyph-by-glyph against the transliteration used throughout.

Folio f2r (Beinecke MS 408), top paragraph. The drawn plant splits each written line into segments. First line, Takahashi EVA: `kydainy · ypchol · daiin · otchal ‖ ypchaiin · ckholsy` (‖ marks where the plant interrupts the line).

The reading is **faithful**. Every word maps to the inked glyphs in reading order; the only discrepancies are of ±1 in *minim* count — the classic ambiguity of whether a run of short vertical strokes is `i`, `ii` or `iii` (e.g. `daiin` vs `daiiin`). These shift a handful of glyph tallies but move no word boundary and no structural statistic. Two corollaries follow. First, the paragraph *self-declares* its section: its prefixes are the herbal set `ch-/cth-/ckh-/sh-`, matching the drawn plant — a single-page instance of the layout–content binding quantified at 165σ in §4.4 and §4.12. Second, this folio carries the very token a well-known 2014 proposal read as a plant name (`kydainy`, its first word); as §4.13 and §6 note, that word is a hapax — it occurs on this folio alone — and cannot be cross-validated, while the positional-locking test of §4 independently excludes the numeric/gematria reading the same first word might invite. The check adds no new result; it confirms that the substrate beneath every result above is sound.

To check that this fidelity is not peculiar to one clean herbal page, we repeated the comparison on two further folios chosen to span the harder cases: `f83r` — the dense biological section, in Currier's *second* scribal hand (B) — and `f1r`, the manuscript's rubbed, faded opening page (whose first line is the specimen reproduced at the head of this paper). Both track the transliteration glyph-for-glyph wherever ink survives; on `f83r` Takahashi's own uncertainty marks fall on exactly the genuinely ambiguous glyphs, and on the rubbed lower half of `f1r` the ink is too faint for glyph-level adjudication — precisely where *any* transcription is least certain. Across the roughly 130 words (~700 glyphs) we could adjudicate over the three folios — spanning both scribal hands, three sections, and a degraded page — we found no substantive error (a wrong glyph or a misplaced word boundary), only the ±1-minim ambiguity, which by the rule of three places a 95% upper bound near 0.4% on the substantive per-glyph error rate of the checked material. This is a spot-check, not a re-transcription: the stronger guarantee remains the near-identical statistics returned by three fully independent transcriptions over the whole manuscript (§4.15).

Appendix B · An illustrative reconstruction (not a decipherment)

§4.16 proves the codebook is undecidable from the text; this appendix makes that *form* tangible. We take the real first line of folio f2r and read it as a filled record. **The glosses in the third column are invented** — chosen to exhibit the mechanism, not asserted: by §4.16 there are on the order of 10^{26} labellings equally consistent with the identical form. The figure shows that the structure *suffices* to carry such content, never what the content is.

WORD (REAL)	PREFIX·CORE·SUFFIX	ILLUSTRATIVE ROLE — INVENTED
`kydainy`	`k·ydain·y`	*entry header / plant name*
`ypchol`	`y·`**`pch`**`·ol`	*a thing + a quality*
`daiin`	`d·aii·n`	*function word ("is / of")*
`otchal`	`ot·ch·al`	*preparation + thing + quality*
`ypchaiin`	`y·`**`pch`**`·aiin`	*the same thing, another quality*
`ckholsy`	`·ckhols·y`	*a thing + a quality*

Folio f2r, line 1, read as a filled record. The third column is **illustrative and invented**; §4.16 shows on the order of 10^{26} alternative labellings fit the same form.

Only one fact here is structural rather than invented: `ypchol` and `ypchaiin` share the core `pch` under different suffixes — the same value in two states, read straight off the line. The template fills without strain, which demonstrates *formal sufficiency*: the machine can carry a materia-medica entry. The glosses are a courtesy to the reader, not a claim — and the pull to believe them is precisely the 97%-false-positive trap of §6.

Method, in brief. Every result in this paper rests on one reusable apparatus: decomposition into prefix·core·suffix inventories; permutation and folio-drift null models behind each association; a statistical-power analysis behind each *negative* result, so that a null bounds an effect rather than merely failing to find one; frontier and local vision models behind the figure-to-code tests; and the rule-space computation of §4.16. The apparatus is deliberately substrate-generic — a detector of whether, and how, a symbol stream carries structured information — and is the study's transferable contribution beyond the manuscript itself.

Companion materials

Self-contained interactive and explanatory companions to this paper:

`The three-dial generator — embedded live.` The generator runs here, inside the paper; a standalone copy is linked below.

- The wheel of f69r — a rotatable reconstruction of the manuscript's own volvelle (§4.13).
- The three-dial machine — a generator that spins the prefix·core·suffix dials on the manuscript's real frequencies and draws each result as glyph forms; a running counter shows how often it lands on a genuinely attested word (§4.13).
- The drifting setting — the suffix drift and text–image decoupling, visualised.
- Findings in three layers — each result as rigorous claim + plain-language gloss + explicit conclusion, with the 25-language comparison.
- Plain-language guide — the argument with every technical term explained by analogy.
- The three-layer map — carries information · a combinatorial machine · the open question.
- Honest self-audit — our contributions vs. prior work, with an action plan for each objection.

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The analyses in this paper — the design and coding of each statistical test, the discipline of holding every result against a control, and the historical synthesis that frames them — were carried out in a single working session in close collaboration with an AI

research assistant (Anthropic's Claude). Candidly, the study would not have been possible, at this breadth or this speed, without it: the tool acted less as an aid than as a force-multiplier on human hypothesis-generation — the difference between a question one could only pose and one that could actually be answered. The direction of inquiry, the interpretations, and responsibility for any errors remain the author's.

References & sources

1. Voynich manuscript — overview and decipherment history. en.wikipedia.org/wiki/Voynich_manuscript
2. Forecasting market: scientifically recognized decipherment by Dec 2026 (~3%, Mar 2026). manifold.markets
3. R. Zandbergen — Transliteration of the Voynich MS text. voynich.nu/transcr.html
4. R. Zandbergen — IVTFF (Intermediate Voynich MS Transliteration File Format). voynich.nu/software/ivtt
5. M. A. Montemurro & D. H. Zanette (2013). Keywords and co-occurrence patterns in the Voynich Manuscript: an information-theoretic analysis. *PLoS ONE* 8(6): e66344.
6. The A/B "language" distinction originates with P. Currier (1976, seminar on the Voynich MS; in M. D'Imperio, ed., *New Research on the Voynich Manuscript*). Quantitatively confirmed by C. Parisel, A Quantitative Confirmation of the Currier Language Distinction (2026), arXiv:2604.25979.
7. Evidence of Layered Positional and Directional Constraints in the Voynich Manuscript: Implications for Cipher-Like Structure (2026). arXiv:2604.19762
8. T. Timm & A. Schinner (2019). A possible generating algorithm of the Voynich manuscript. *Cryptologia* 44(1): 1–19. (The "self-citation" auto-copy generator.)
9. G. Rugg (2004). An Elegant Hoax? A Possible Solution to the Voynich Manuscript. *Cryptologia* 28(1): 31–46. (The table-and-Cardan-grille hoax method.)
10. A new study suggests the Voynich Manuscript may be a medieval cipher (Jan 2026). archaeologymag.com
11. On the combinatorial-wheel lineage: R. Llull, *Ars Magna* (c. 1305); L. B. Alberti, *De Cifris* (c. 1467) — the first polyalphabetic cipher disk, ancestor of the rotor machines. Alberti cipher · rotor lineage
12. W. F. Friedman's conclusion that the Voynich is a synthetic "a priori" language — History of research. voynich.nu/solvers.html
13. S. Reddy & K. Knight (2011). What We Know About the Voynich Manuscript. *Proc. 5th ACL-HLT Workshop on Language Technology for Cultural Heritage, Social Sciences, and Humanities*, pp. 78–86. aclanthology.org/W11-1511. (Character entropy, morphology, n-gram structure.) The low-entropy anomaly also in C. Bennett (1976), *Scientific and Engineering Problem-Solving with the Computer*.

14. J. Stolfi (1997–2005). Voynich manuscript — word structure and the "crust–core–mantle" decomposition; statistical notes. ic.unicamp.br/~stolfi/voynich
15. J. Bowern & L. Lindemann (2021). The Linguistics of the Voynich Manuscript. *Annual Review of Linguistics* 7: 285–308. (Authoritative survey of the statistical and linguistic evidence, incl. LAAFU / line-as-unit and "Grove words".)
16. Foundational: C. E. Shannon (1948), *A Mathematical Theory of Communication*; G. K. Zipf (1949), *Human Behavior and the Principle of Least Effort*; M. A. Montemurro & P. A. Pury (2002), long-range correlations in literary corpora.

Data: Takahashi transliteration (36,923 tokens), Landini–Stolfi IVTFF archive. Baseline: `Don Quijote` (Project Gutenberg). All statistics computed on the full corpus. This draft reports an authenticity assessment only and makes no claim of decipherment.